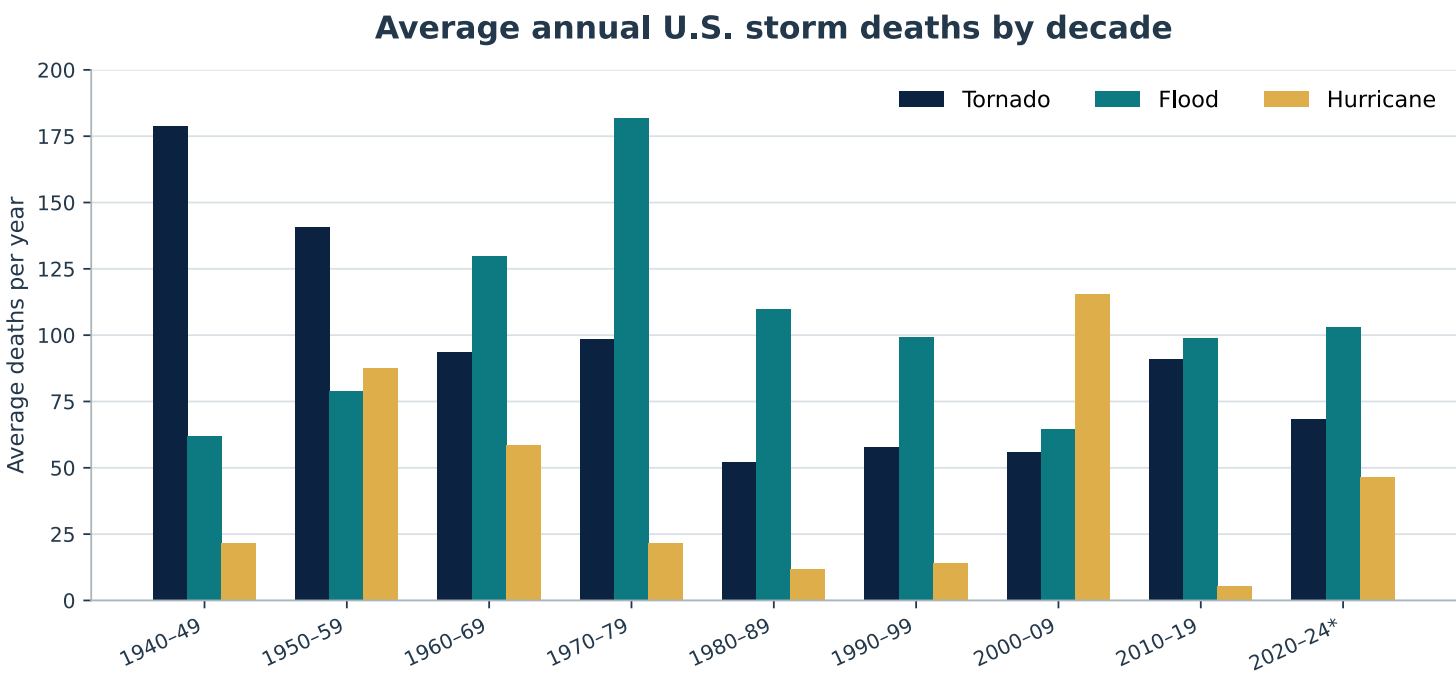


More People in Harm’s Way, Fewer Storm Deaths per Person.

A PLAIN-ENGLISH LOOK AT U.S. STORM RISK, EXPOSURE, AND ADAPTATION

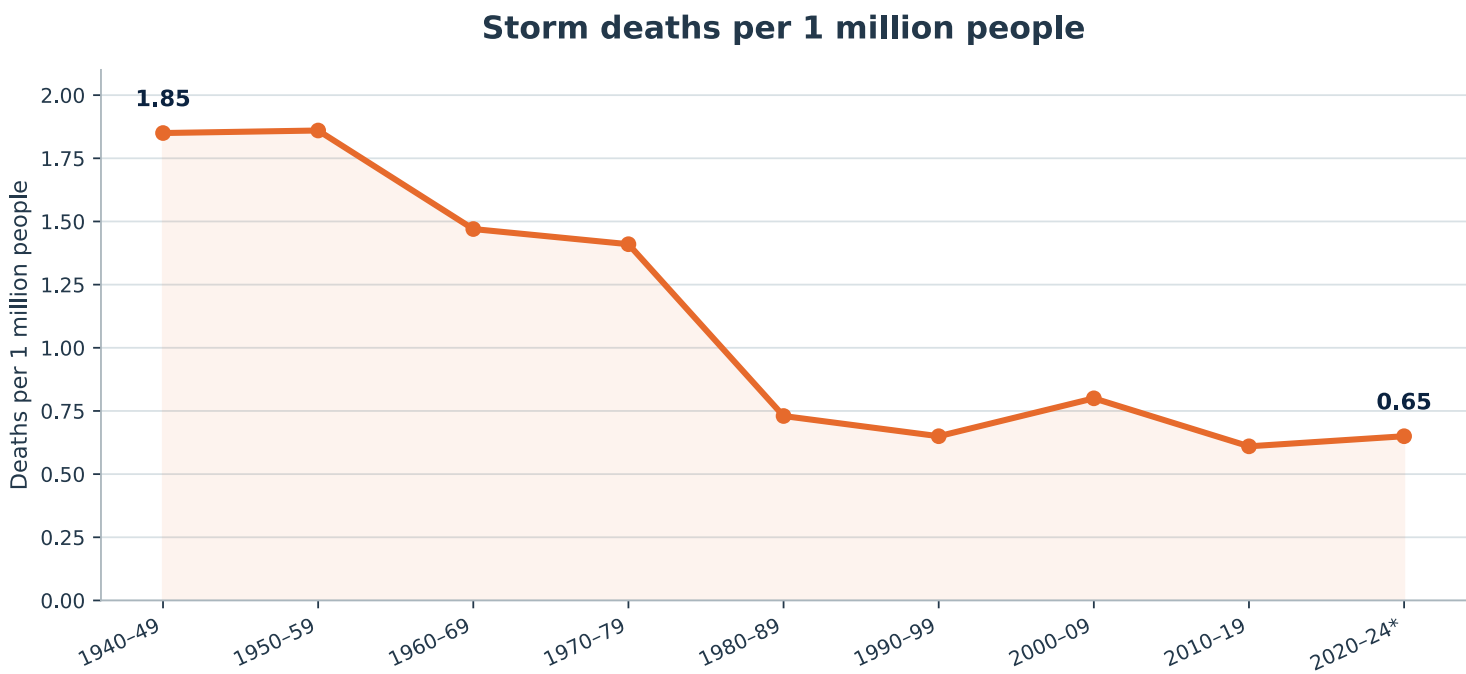
Official U.S. data show that deaths from tornadoes, floods, and hurricanes have generally gone down over time—especially after adjusting for population. At the same time, more people live in hurricane-prone coastal areas. That pattern fits the idea that Americans have gotten better at protecting themselves, but it does not prove adaptation is the only reason.

1 Storm deaths went down over time



Deaths generally fell after the 1970s, though especially bad disasters can push up a decade.
NOAA/NWS Weather Related Fatality and Injury Statistics. 2020–24 is partial.

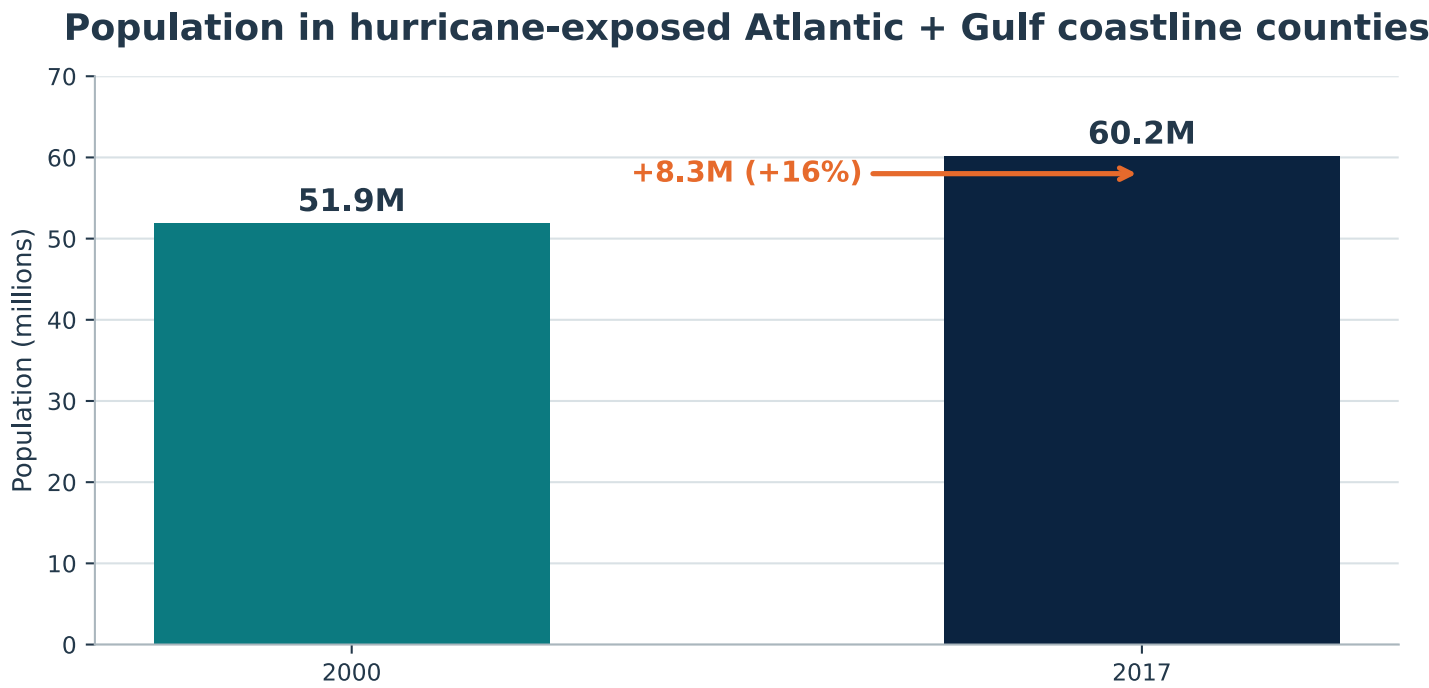
2 The drop is bigger when we account for population



About 65% lower per person: from about 1.85 deaths per million in the 1940s to about 0.65 in 2020–24.

Population: U.S. Census Bureau.

3 More people live in hurricane-prone coastal areas



In 185 Atlantic and Gulf coastline counties most exposed to hurricanes, population rose from 51.9M in 2000 to 60.2M in 2017 (+16%).

Important: population growth is not the same as domestic migration. Births and international immigration also matter.

U.S. Census Bureau, Coastline America (2019).

What this likely means

More people now live in risky coastal areas, but storm deaths—especially per person—are lower. That fits the idea that forecasting, evacuation, building standards, emergency response, behavior, and wealth have improved protection. It still does not prove exactly why deaths fell.

Economics: adapting may cost less than we think

- \$600B vulnerable structures (assignment estimate)
- 50-year life → about \$12B replaced each year
- 15% safer-location premium → about \$1.8B extra each year
- ≈0.0055% of a \$33T economy

Illustrative classroom calculation, not a forecast.

But wait

Population growth is not the same as migration. Moving south does not always mean moving into a hurricane zone. Fewer deaths also do not prove climate risk itself is smaller.

IPCC: exposure drives losses

SREX says long-run disaster losses rise largely because more people and valuable property are in harm’s way, along with vulnerability—not simply because the hazard changed.
IPCC, SREX (2012).

Heat is a different story

The 1995 heat-death spike shows that age, social isolation, access to air conditioning, and emergency response can strongly shape mortality.
CDC / NWS.

4 A different metric: Burke et al. (2025)

Burke and coauthors estimate temperature-attributable mortality using statistical relationships between temperature and deaths. That is much broader than NWS counts of deaths directly recorded under weather-event categories, so the two numbers should not be treated as the same measure.

Sources

1. NOAA/NWS. Weather Related Fatality and Injury Statistics — weather.gov/hazstat
2. U.S. Census Bureau. historical population data — census.gov
3. U.S. Census Bureau. Coastline America (2019) — census.gov/library/visualizations/2019/demo/coastline-america.html
4. IRS Statistics of Income migration data — irs.gov/statistics/soi-tax-stats-migration-data
5. IPCC, SREX (2012) — ipcc.ch
6. Burke et al. (2025). NBER Working Paper 34313 — nber.org/papers/w34313
7. CDC. 1995 Chicago heat wave — cdc.gov

Method note

Death intensity = average annual tornado + flood + hurricane deaths per 1,000,000 people. 2020–24 is a partial decade. NWS counts hurricane wind deaths separately from flood and tornado deaths.